

A Bayesian Framework for Morphological Profiling: Phenotype Clustering, Pathway Decomposition, and Gene Discovery

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Abstract

Chimeric antigen receptor (CAR) T-cell therapies show promise for treating hematologic cancers, but their efficacy and accessibility are limited by the difficulty of experimentally screening large numbers of CRISPR perturbations in primary T-cells. CRISPR screens have identified gene knockouts like ACAT1 that enhance T-cell function, but these screens are expensive and technically difficult to perform in primary T cells. We developed a computational approach to prioritize candidate gene knockouts using morphological profiling data from the publicly available JUMP Cell Painting dataset, which contains high-dimensional imaging features for thousands of CRISPR knockouts in non immune cells. Using ACAT1 as a biological anchor, we applied fuzzy k-means clustering to identify genes with similar morphological signatures and implemented ridge regression models to decompose gene effects into pathway-level and gene-specific components. We compared two decomposition strategies: one based on gene ontology (GO) annotations and another based on morphological clusters. The pathway-based model identified candidates within biologically relevant pathways shared with ACAT1, while the cluster-based model prioritized genes producing similar morphological changes regardless of pathway membership. Both models explained meaningful variance in the data (14.9-16.4% R^2) and identified biologically plausible candidates including KIF2B and NDUF57, genes involved in cytoskeletal dynamics and mitochondrial metabolism that could potentially enhance T-cell function. Our approach demonstrates that morphological profiling in non-immune cells can serve as a scalable screening tool to prioritize gene targets for experimental validation in CAR-T engineering, reducing the cost and complexity of discovering therapeutically relevant gene knockouts.

Background

In the realm of emerging cancer therapies, CAR-T cell therapy stands out for its potential to revolutionize treatment of leukemias, lymphomas, and multiple myelomas [1]. Despite CAR T-cell therapy's clinical efficacy, several obstacles still remain: its complex manufacturing, non-universality across patients, and the high risk of side effects such as cytokine release syndrome (CRS), and immune cell-associated neurotoxicity syndrome (ICANS) [2]. CRISPR-based genetic screens have emerged as a powerful tool to systematically optimize CAR-T cell therapies by identifying gene targets that enhance T-cell persistence, activation, metabolism, and resistance to immunosuppression. Several landmark studies using this approach have uncovered critical intracellular immune checkpoints, whose knockout restores T-cell activity and enhances tumor clearance in preclinical models. One such checkpoint modulation is knockout of the ACAT1 cholesterol esterification enzyme, which reprograms cellular metabolism in T-cells to increase membrane cholesterol levels, leading to a heightened antitumour response in engineered cells [3]. CRISPR screens to reveal new KO targets for enhanced tumor clearance are promising but remain limited by cost, scalability, and the difficulty of performing pooled selection screens in primary T cells, which are notoriously hard to edit and culture. This creates a need for complementary computational or phenotypic approaches to prioritize gene candidates before undertaking expensive and technically challenging immune-based validation experiments.

Morphological readouts from high-content imaging can act as a proxy for cellular state changes, reflecting shifts in cytoskeleton, organelles, nuclear architecture, and metabolism [4]. The publicly available JUMP-CP dataset includes thousands of CRISPR KO perturbations with detailed morphological feature profiles, offering an unprecedented opportunity to repurpose non-immune cell imaging to infer candidate gene targets for T-cell engineering[5]. While previous research has demonstrated that morphological features correlate with mechanisms of action for small molecules and chemical perturbations, there is a notable gap in applying these phenotypic signatures to immune functional engineering [6]. This project will improve scientific knowledge by bridging morphological phenotyping with immuno-engineering: we will build a machine learning classifier that learns from known T-cell-beneficial KO signatures such as ACAT1, and applies it across the entire KO morpheme to rank novel candidate genes. This will provide a novel computational screening strategy that dramatically lowers the barrier to T-cell engineering; by leveraging existing data rather than expensive new screens, it enhances capability and accelerates discovery. Should this methodology identify robust candidates whose knockout improves T-cell

persistence or cytotoxicity, it could directly translate into engineered T-cell therapies with improved efficacy and lower cost.

To incorporate biological structure into the analysis of gene-mediated morphological effects pathway and functional annotation databases provide a principled means of aggregating gene perturbations into higher-order biological programs. The Kyoto Encyclopedia of Genes and Genomes (KEGG) organizes genes into manually curated pathways representing canonical metabolic and signaling processes, offering high interpretability and mechanistic specificity. In parallel, the Gene Ontology (GO) dataset organizes genes according to hierarchically structured annotations describing biological processes, molecular functions, and cellular components, allowing functional characterization across multiple biological scales [7, 8]. Together, these annotation frameworks provide complementary biological structure for organizing high-dimensional gene perturbation data and facilitate interpretation of morphology-derived signatures in terms of known cellular processes and regulatory programs.

Results and Discussion

Clustering

To explore gene perturbations, from a morphological perspective, we used the JUMP Cell Painting (JUMP-CP) dataset, which contains standardized morphological profiles for thousands of CRISPR knockouts. Each gene knockout is represented by a high-dimensional feature vector which quantifies organelles including nuclei, cytoplasm, and mitochondria. These features capture texture, granularity, intensity, and shape metrics, providing a phenotypic snapshot of cellular state after perturbation. Because morphological readouts integrate changes across multiple biological processes such as metabolism and stress responses, they offer an opportunity to identify patterns that may generalize across cell types, including those relevant to immune function.

Why Fuzzy K Means

We applied fuzzy k-means clustering to test whether morphological profiles yield biologically interpretable groups. Unlike standard k-means, fuzzy k-means assigns membership probabilities across clusters rather than forcing hard assignments. This is biologically appropriate because gene knockouts often influence multiple cellular processes simultaneously, so a perturbation may partially resemble several morphological states rather than belonging cleanly to just one. Across various values of k , clusters corresponded to recognizable morphological themes such as mitochondrial texture changes, increased ER or RNA granularity, and shifts in cytoplasmic intensity.

Fuzzy k-means can be understood through the Expectation–Maximization (EM) framework. In the E-step, the algorithm estimates the probability that each gene belongs to each cluster based on the current cluster centers. In the M-step, it updates cluster centers by taking weighted averages of gene profiles, using the membership probabilities as weights. This iterative soft assignment allows the algorithm to identify shared morphological characteristics while preserving the reality that perturbations may span multiple biological pathways. The EM interpretation also makes the output naturally compatible with downstream probabilistic modeling, because cluster memberships can be treated as latent variables representing cellular states.

After computing fuzzy membership scores, we examined how different gene knockouts distributed across clusters. Genes did not collapse into a single dominant cluster; instead, they showed distinct but partially overlapping membership patterns, with some clusters enriched for genes annotated to lipid metabolism and membrane-associated processes. These trends are consistent with ACAT1's known role in cholesterol esterification and membrane remodeling. Using ACAT1's cluster membership vector as a reference, we identified genes with similar membership patterns using cosine similarity and KL-divergence metrics. Many of the closest neighbors of ACAT1 included genes involved in lipid metabolism, intracellular trafficking, and other membrane-proximal pathways, suggesting that morphological similarity can reflect functional relationships. Several genes showed moderate membership across multiple clusters, which is biologically plausible given that metabolic perturbations often have secondary effects on stress response, organelle organization, and mitochondrial function. Overall, the clustering results demonstrate that the morphological structure in JUMP-CP is interpretable and biologically meaningful.

ACAT 1 Analysis

We first tested our approach on a small preliminary dataset consisting of only 5 ACAT1 knockout samples and 10 wild-type control samples. We ran the fuzzy k-means model both with and without PCA on this subset. In all cases, ACAT1 and control samples showed only modest differences in their latent-state probabilities and substantial overlap in feature space. For example, with PCA, as shown in Figure 1, ACAT 1 knockout cells had a higher probability of occupying state_0 (0.571) compared to control (0.312), while controls were more likely in state_1. When the model was run without PCA, both ACAT1_KO and control cells had a higher probability of being in state_1 (0.557 and 0.650, respectively). This suggested that, at this small sample size, we could not confidently claim that ACAT1 occupies a clearly distinct morphological state; the signal was suggestive but weak, and sensitive to modeling choices.

Latent-state distributions by condition:

	state_0_prob	state_1_prob
Condition		
ACAT1_KO	0.571050	0.428950
control	0.312837	0.687163

Figure 1: When PCA is used, ACAT1 Knockout cells are more likely to be in state 0, while control cells are likely to be in state 1. However, individual samples have measurable overlap.

To obtain a more robust picture, we then used the full JUMP-CP dataset and passed it through a preprocessing pipeline before clustering. In this pipeline, we aggregated Cell Painting features to the gene level by averaging all profiles for each gene, computed gene effects relative to wild-type by subtracting the WT mean feature vector, removed low-variance and highly correlated features to reduce noise and redundancy, imputed any remaining missing values, and standardized each feature to zero mean and unit variance. We then applied PCA to this cleaned, WT-centered gene $\times$ feature matrix to obtain a low-dimensional latent representation, and ran fuzzy k-means on the resulting gene-level latent space.

To move beyond a single-gene comparison and explore structure across all perturbations, we applied the fuzzy k-means model to the full gene-level latent space and set $k = 40$ clusters. This produced, for each of the 388 genes, a 40-dimensional probability vector describing how that gene's WT-centered morphology is distributed across the 40 latent clusters. We then used these gene-level probability distributions to look for commonalities between gene types. In particular, we hoped to determine whether genes with related functions or pathways tend to share similar cluster membership patterns.

Figure 2 below shows that cluster 28 contains the largest number of genes, meaning that for more genes than any other cluster, cluster 28 is the highest-probability latent state in their fuzzy membership profiles. In other words, when the model is forced to pick a single "most representative" morphological state for each gene, many genes are best described by the state represented by cluster 28. The gene list for this cluster includes transporters such as ABCA12 and ABCB5, metabolic enzymes such as ACADVL, FAAH, LPCAT1, and UGCG, and trafficking or signaling genes like RAB11B, CD244, and SYT1. This mix suggests that cluster 28 may capture a morphology associated with membrane remodeling, vesicle trafficking, and lipid or energy metabolism, which are processes that are broadly relevant to how cells respond to perturbations. While fuzzy memberships mean these genes can still participate in other states, their shared dominant assignment to cluster 28 indicates a common underlying morphological theme that the model consistently recognizes across diverse perturbations.

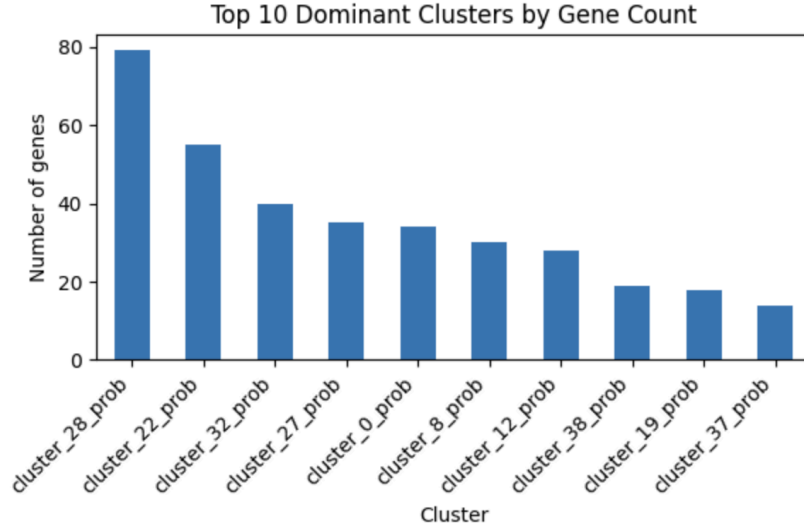


Figure 2: Depicts the ten most dominant clusters out of the 40

It is also important to note that cluster 0 is the dominant cluster for ACAT1 and a set of genes that are broadly involved in metabolism and signaling, including enzymes and transporters (e.g., ALDOC, CYP4B1, COX7A2, PPT2, SLC22A18, SLC38A4) as well as growth- and matrix-related factors such as TGFB1, CTGF, and MSTN. The fact that ACAT1 groups with genes linked to energy metabolism, lipid handling, and extracellular signaling suggests that the morphology captured by cluster 0 reflects a coordinated change in metabolic and signaling state rather than a purely structural phenotype. This is consistent with ACAT1's role in cholesterol esterification and membrane remodeling, and supports the idea that its morphological signature is embedded within a broader metabolic–signaling context rather than being an isolated, idiosyncratic effect.

Gene-pathway decomposition through ridge regression

Since morphological phenotypes are cumulative outcomes of individual gene effects [9], we hypothesized that augmenting our modeling with biological pathway information for every gene knockout would enable stronger inference and predictive power. To do this, we decompose the morphological effect from each gene (g) knockout, Θ_g , into a pathway-level morphological effect, Ψ , and a gene-specific effect δ_g . This parameterizes a generalized linear model (Equation A) that can then be fit through ridge regression.

$$\Theta_g = A_g \Psi + \delta_g \quad (\text{A})$$

This approach encodes our prior beliefs about the pathway effects in two key ways. First, the structure of the linear model explicitly filters gene-level effects through their pathway memberships, formalizing our assumption that genes within the same pathway produce correlated morphological changes. This structural assumption parallels the confirmatory factor analysis framework of Pham et al. (2016), who modeled gene expression as a function of latent biological pathway factors, constraining genes to load only onto their corresponding pathways [10]. Second, the ridge penalty provides regularization informed by biological structure. Zeng et al. (2021) demonstrated that incorporating external meta-features like pathway membership or functional annotations *a priori* into penalized regression yields improved prediction performance and interpretability [11]. In their framework, meta-features determine feature-specific penalty parameters for gene expression data, while in ours, λ encodes a prior belief about the strength of pathway-level morphological effects relative to gene-specific deviations.

Biological pathway information for every gene in the Cell Painting dataset was collected from the Kyoto Encyclopedia of Genes and Genomes (KEGG) database and formulated into the membership matrix, A . However, because KEGG is limited to expert labeled pathway maps, only 216/388 genes had pathway representations and the average genes/pathway (G/P ratio) was only 8.3, with only 1.8 pathways/gene on average. This resulted in poor model performance in initial testing. Therefore, we turned to the Gene Ontology (GO) database which has higher gene coverage, with broad functional characterization rather than membership in canonical pathways [7, 8].

Although the GO database is less curated as KEGG, it still allows us to generalize beyond individual gene behavior, and we reasoned that the benefits of higher coverage would outweigh any noise introduced. Querying GO terms for our Cell Painting represented genes gave 86% coverage compared to 64% coverage with KEGG with more than double the gene-label pairs. With stricter thresholds, the average genes/pathway increased to 15.3.

Cell-level features were extracted from the Cell Painting dataset and gene effects were calculated through maximum-likelihood estimation by averaging across replicate wells. An average was also calculated across all control wells with no gene knockout (WT) which was subtracted from the gene averages to get an average effect vector, Θ_g , across all features for every gene knockout. Features with variance below $1e-4$ ($n = 947$) were filtered out to remove non-informative features. One gene from every pair of features with covariance above 0.9 was also removed ($n = 2521$) to limit redundancy and further reduce dimensionality. Features were then normalized to zero-mean and unit variance using the Scikit library in Python. This resulted in a final gene effect matrix, Θ , with 248 genes and 205 features.

The pathway effects were then found by fitting the data to Equation A through ridge regression. Learning from labeled data with prior information makes this problem MAP estimation (Equation B). With an uninformative prior on δ , $P(\delta) = 1$, this becomes a simple optimization with one penalty (Equation C), and the gene-specific effects become the residuals after fitting (Equation D).

$$\hat{\Psi}, \hat{\delta} = \arg \max P(\Psi, \delta | \Theta, A) \quad (B)$$

$$= \arg \max P(\Theta | \Psi, \delta, A) P(\Psi) P(\delta)$$

$$\hat{\Psi} = \arg \min_{\Psi} ||\Theta - A\Psi||^2 + \lambda ||\Psi||^2 \quad (C)$$

$$\hat{\Delta} = \Theta - A\Psi \quad (D)$$

The model was optimized using leave-one-gene-out cross validation (CV) measure mean square error and by comparing performance in CV to a pathway-naïve model for gene effect prediction. By constraining the number of genes per pathway to 10, the model performed optimally with pathways explaining 14.9% of morphological variance (R^2) and a 4.2% improvement over a pathway-naïve model in CV. Lowering the G/P threshold to 2 improved R^2 to 30.2%, but also decreased the improvement, indicating the low threshold allowed overfitting. We additionally selected an optimal penalty parameter λ through CV. The optimal λ was found to be 20 (Figure BA) which indicates modest regularization improves generalization without shrinking pathway signals too much. Top features for each GO Process (Figure BB) show that the model captures expected biological effects. For example, the top feature for the Translation process is Endoplasmic Reticulum granularity, a predictable effect of ribosome activity.

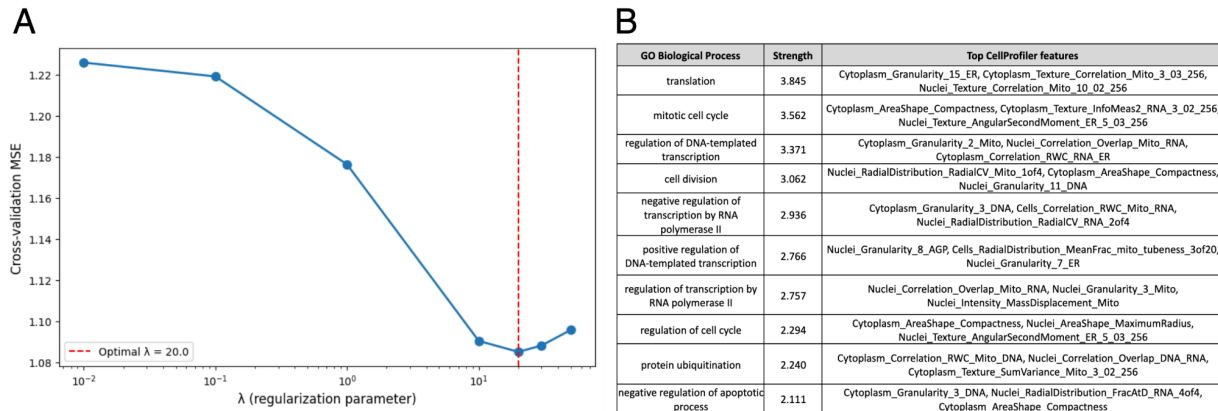


Figure B. Pathway model evaluation. (A) Cross-validation across λ values. Leave-one-out cross-validation was calculated for the model with varying penalty parameters λ which control the regularization of pathway values during the ridge regression. The U shape indicates the optimal value found is a true optimum. An optimal λ of 20 indicates that moderate regularization improves generalization with reduced pathway strength. (B) Top related features for GO Processes. The top biological processes by strength, or magnitude of the pathway vector, were ranked and their top features by magnitude are listed. The results exhibit biological significance with endoplasmic reticulum granularity being a top feature for translation.

Querying ACAT1 against all other genes through cosine similarity between pathway signature and gene-specific effect allows ranking of potential gene knockout candidates to produce similar effects. The pathway similarity ensures biological plausibility of candidates. With better pathway coverage and annotation, these results could be improved to indicate more unique relationships between genes.

Table A. Gene query results for ACAT1. Model representation for genes from the Cell Painting dataset were compared to ACAT1. The Score was found by calculating the average cosine similarity between the pathway components (Ψ) and the gene-specific components (δ). Pathway similarity of 1.0 indicates the query and candidate gene were represented by only the two shared pathways.

Gene	Score	Gene-specific (Residual) Similarity	Pathway Similarity	Shared Pathways
ACSL6	0.629	0.258	1.000	lipid metabolic process; fatty acid metabolic process
CYP4F2	0.618	0.236	1.000	lipid metabolic process; fatty acid metabolic process
FADS3	0.612	0.224	1.000	lipid metabolic process; fatty acid metabolic process
LYPLA1	0.572	0.144	1.000	lipid metabolic process; fatty acid metabolic process
ACOT6	0.554	0.333	0.774	fatty acid metabolic process
ACOT11	0.552	0.244	0.859	lipid metabolic process; fatty acid metabolic process
OLAH	0.522	0.044	1.000	lipid metabolic process; fatty acid metabolic process
ECI1	0.521	0.043	1.000	lipid metabolic process; fatty acid metabolic process
CYP2J2	0.467	-0.066	1.000	lipid metabolic process; fatty acid metabolic process
CYP4B1	0.465	0.156	0.774	fatty acid metabolic process

Gene-cluster decomposition through ridge regression

By clustering the gene knockout feature data using a fuzzy KMeans algorithm, we were able to assign probabilistic gene memberships to biologically relevant morphological groups. Now, we can compare the biologically relevant pathways data with the clustering data by replacing the A matrix with the soft counts. In the JUMP feature dataset we used, 128 pathways were found in the GO database, with an average of 2.8 pathways membership per gene. Thus we decided to create 40 clusters, modeling each cluster as a 'pathway'. While each gene would technically have memberships to each cluster, the cluster effects (previously pathway effects) would be scaled by its probability to be in said cluster. Other than switching the A matrix, we kept all other parameters the same for a proper comparison.

To identify similar gene candidates, we compared their pathway similarity and their individual similarity to ACAT1. As our model gene knockout, ACAT1 is part of multiple metabolic pathways, but also has its own effect directly affecting cell morphology through cholesterol in the cell membrane. We theorized that many genes may be part of a biologically relevant pathway, but they did not possess a strong individual effect on the cells themselves. Therefore, we conducted our comparison to attempt to find genes which had strong pathway effects, but still had strong individual differences. This ensures that our selected genes are not all strong candidates just because they are on the same pathways (Ψ_g) as ACAT, as knocking out multiple genes from the same pathway may have diminished effects, but knocking out their own individual effects (δ_g) as well, maximizing the knockout efficiency. Again, the ridge regression model is significant, explaining 16.4% of clustering morphological variance (R^2), and a 15.2% improvement over a cluster-naive model.

Table B. Top 10 gene query results for ACAT1, identifying candidate genes from the JUMP CP dataset using associated cluster probabilities from the fuzzy KMeans algorithm. Higher score indicates a closer similarity between the gene and ACAT1.

Gene	Score	Gene-specific (Residual) Similarity	Cluster Similarity
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KIF2B	0.727	0.619	0.836
PHLPP2	0.714	0.486	0.941
CHD6	0.713	0.439	0.986
BHMT2	0.711	0.426	0.996
NDUSF7	0.708	0.479	0.936
BCO1	0.693	0.483	0.904
MSTN	0.689	0.400	0.978
RRAGD	0.681	0.411	0.952
ACOT6	0.677	0.359	0.995
RGS17	0.673	0.370	0.976

To compare the two results, it appears that the pathways based data inherently weights pathway effects higher than the gene specific effects, while the clusters based predictions weight morphological feature similarity and individual gene functioning. Pathway memberships are binary variables, meaning that if a gene is outside of ACAT1's (or the query gene's) pathways, it will have a low pathway similarity and reach a low score even with a high individual similarity. This indicates that pathway predictions allow us to discover similarly strong individual effects within biologically relevant pathways. On the other hand, the cluster-based ridge regression presumes more about the biological relevance of each morphological cluster, allowing genes which produce similar individual effects to show more prominently. A clear example of this is the KIF2B gene. Ranked 52nd under the pathways-based regression model, it became the highest ranked gene knockout under the cluster based model for its high gene-specific effect. When comparing candidate results between the pathways and the clusters, it is important to note that the cluster results have higher scores overall and align more closely with the JUMP CP perturbation match similarity ranking. We predict this is because the clusters themselves are based on the original cell morphology feature data which the similarity ranking is also based on.

Upon further examination of high-scoring cross-method hits, we note that some of the found genes mediate potentially relevant biological pathways. One such hit: KIF2B, is involved with microtubule motor activity and could potentially be an effector of T-cell motility [12]. Because cytoskeleton-mediated T-cell migration is a fundamental aspect of its cancer-killing function, perturbation of the KIF2B gene could be a valuable hit to examine further [13]. Interestingly, this gene shares no gene path similarity with ACAT1, but has a highly similar cell morphology readout, potentially indicating a link between membrane cholesterol levels as mediated by ACAT1 and cell migration. Another significant hit could be NDUSF7, a metabolic signal which forms the mitochondrial respiratory chain [14]. Like with ACAT1, perturbation of the NDUSF7 gene could potentially impair oxidative phosphorylation and in turn, alter the cellular metabolic state, which could lead to higher TCR clustering like with perturbation of ACAT1, or affect other relevant metabolic pathways [15]. Measuring tumor cell killing, cytokine secretion, and T-cell persistence in this context would provide a direct readout of whether the predicted gene knockouts enhance antitumor T-cell responses.

Conclusion and Future Goals

In conclusion, we have demonstrated two related but different morphological-feature based regression models used to decompose gene knockout feature data into its batch-specific and individual-specific effects. These models were based on the same feature data, but performed differently based on the gene associations provided. The pathways-based model identifies other candidates within the same or similar pathways as the query gene which also have a similar individual morphological impact. This model would be most useful when attempting to optimize the knockout possibilities within a specific biologically relevant pathway. If given a gene within a significant pathway which cannot be knocked out easily, the model will identify gene knockouts within the same pathways which can have variable individual effects. This allows the user to identify a pathway, and choose between which type of individual morphological change they would prefer, either one which focuses only on the pathway, or one which has additional functions. The clusters-based model is more useful when identifying genes

which have similar gene-specific effects on cell morphology. If given a specific useful morphological change, this model can identify other gene knockouts which create the desired morphological profile, without restricting the biological pathway. In the case of ACAT1, we highlighted KIF2B as a candidate gene which created a similar morphological profile, but is not on any of the same cellular pathways. Overall, we hope that these models can serve as a generalizable and scalable tool for any target query gene with morphological effects, allowing for better informed CRISPR knockout research within the CAR T-cell field, or any other cellular engineering applications.

Future improvement of the pathway model could implement full bayesian optimization with Gibbs' sampling to find the true optimal contributions of pathway- and gene-level effects. Uncertainty estimations with full Bayesian modeling would also provide more useful predictions for gene finding. Additionally, creating a refined Cell Painting dataset composed of genes with many pathway associations would give better coverage and show the maximum potential of the pathway decomposition approach.

To truly assess the relevant effects of these CRISPR KO perturbations in mediating CAR T-Cell efficacy, further studies would need to be conducted in vitro. For this, relevant guide RNAs would be designed to knockout hit genes and delivered to T-cells as a ribonucleoprotein complex with the CRISPR Cas protein [1]. Co-culture killing studies such as a B16-F10 melanoma-Pmel-1splenocyte study, could elucidate the efficacy of edited cells in eliciting a heightened T-cell response [16].

Comparison to Proposal

In the initial project proposal, a highly generalized version of the current algorithm was outlined. Its conceptual framing aimed to link knockout signatures of T-cell beneficial CRISPR KOs, such as SOCS1, ST3GAL, PRODH2 [1]. It thus used the general assumption that any given immune-relevant perturbation would have transferable morphology with non-immune cells. The reviewers correctly identified a notable weakness with this framing: the lack of mechanistic grounding for the link between immune signaling and metabolic genes and morphological changes. In subsequent revisions we grounded the approach in an anchor gene: ACAT1, whose role in cholesterol metabolism directly affects cell membrane composition and provided a clearer rationale for observing morphological effects in Cell Painting assays. This revision also addressed reviewer concerns by clarifying feature selection, adding batch normalization strategies, and reducing reliance on unsupported distributional assumptions.

The final report largely fulfills the proposal objectives while also introducing new methodology that naturally emerged during algorithm implementation. In addition to implementing probabilistic clustering of morphological phenotypes derived from the JUMP-CP dataset, we also opted to observe pathway and cluster-level ridge regression models to assess the shared effects of biological structure and gene-specific residuals. Due to poor initial performance of the KEGG dataset as a result of poor pathway coverage and interpretability, the final report shifted its focus from KEGG to GO annotations, which improved model stability and interpretability. Although the final model does not fully achieve the original proposal's broad generalization across many T-cell-beneficial knockouts, this reflects a deliberate reduction in hypothesis space rather than incomplete execution.

Commentary on experience

Looking back on our team's experience, we thought that this project has been beneficial to our understanding of a computational biology research project, and now understand a bit better how to approach such a project. The first barrier we encountered was choosing a dataset. There is a vast amount of data online, but it is difficult to access or find biologically relevant data if you are unsure of your project idea itself. We were initially attempting to find a dataset, then come up with a novel application, or trying to fit a problem to a dataset. In our case, we had many datasets and project ideas to choose from, but were having difficulty trying to mold projects onto a dataset, or vice versa. This explains the disconnect between the CRISPR KOs and the JUMP dataset we had in our first project proposal. Going back, we would instead have a project idea and then find corresponding data for said issue. We also ran into trouble with finding the actual csv files themselves, and next time would ensure that we had the raw data before committing to a project.

We believe that we managed our time well, balancing approach execution and report writing. Furthermore, we were able to assess which aims would take the most time, and allotted accordingly. Surprisingly, the most difficult aspect of the execution of this project would be communicating with each other. After finding relevant datasets and creating polished code it is easy to share what they do, but it is difficult to keep track of who has looked where for data, and to interpret each others' code. Sharing code was the most difficult, as it would require a lengthy explanation of personal stylistic coding choices to fully understand each other, and making even simple edits could have lengthy ripple effects across the script when not fully informed. In the future, it would be beneficial to keep more consistent communication lines, ensuring that everyone is on the same page, and allowing for less confusion on project progression.

Although it was a difficult and challenging process, there were moments which made this project satisfying. A top moment for our team was when we were looking for the correct feature data. The JUMP CP dataset is large, and navigating such a large dataset proved difficult. We had to comb through metadata to find where to even look for our raw data, and then had to parse raw data to ensure that the correct gene knockouts and features were included. The moment when we became familiarized with the dataset and could locate the ACAT1 data was the moment when we realized that our project was actually feasible. Another rewarding moment was our discussions on how to approach model design. To include more biologically relevant parameters in our model we first thought of using ridge regression with the pathways (our aim 2), but then we had to think about how to combine that approach with the clustering algorithm we were already building for the JUMP CP dataset (our aim 1). Through continued discussions, we eventually came up with our aim 3, where we could use the clustering as our 'pathway' assignments, bridging the two ideas into biologically relevant models. This project had its bumps and uncertainties, but the challenge allowed for a fulfilling conclusion, creating a tool which we believe could be truly useful.

Commentary on peer-review process

We felt that the peer-review process was mildly helpful, encouraging us to work on our proposal and giving us another perspective on project writing. The comments posted on our proposal were generally issues we already recognized, but having to write a response and think of an alternative approach helped promote creative ideation on our project. Otherwise, the comments were not extremely eye-opening or useful, as they did not provide insight into any initially unrecognized direction or pitfalls. It also did not help that our feedback consisted of two seemingly AI generated responses and 10 minor comments on our proposal, such that our feedback was more factual instead of meaningful perspectives. Our team benefited much more when we did the reviewing ourselves. Of note was one proposal which was focusing too much on the biological relevance of their unmet need, that they had not recognized what they were attempting to build was too specific and was already designed. In turn, we were able to widen the scope of our project. We were no longer testing for specific CRISPR knockouts, but were instead building a useful and generalizable tool within the context of a biological application. This newfound view on project validity and applicability allowed us to explore further beyond the JUMP CP dataset and look further into the novel idea of splitting between pathway and individual morphological changes.

Division of Labor

J.K. developed the approach, code, and wrote up results and future directions for the ridge regression and pathway effects decomposition.

M.S. developed the core project concept of applying Cell Painting-derived morphological profiling to CAR T-cell engineering, wrote the biological background, analyzed and wrote about the results and future directions within a biological context, and wrote the comparison across proposals section.

A.M. developed the approach and code for the fuzzy k-means-based morphological clustering and gene-level latent state distributions, and contributed to the write-up of the clustering results and their interpretation.

M.C. related the morphological relevance of CAR T-Cell gene knockouts using the ACAT1 gene, located feature data, merged the k-means results to execute within the ridge regression code to write results for gene-cluster regression and the model applications in the conclusion.

S.M. identified the JUMP-CP dataset, conducted initial data exploration, extracted ACAT1-related perturbation data from the full dataset, and wrote the morphological signature results and abstract.

Use of AI Tools

Claude was used to assist with the literature search for related approaches to the pathway-decomposition and formalization of the regression analysis. It was also used to assist with implementation of the regression analysis code. Scripts to query information from the KEGG API and GO database were mostly written with Claude, as these tools were unfamiliar to the group. Otherwise, it was used for output text formatting, debugging, and optimization when needed.

Code Availability

Code is available in the ZIP file associated with this report.

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